

Lecture 10: Relations

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1 Binary Relations

Let

$$A = \{a, b, c\}, \quad B = \{1, 2, 3\}.$$

Their Cartesian product is the full set of nine possible ordered pairs:

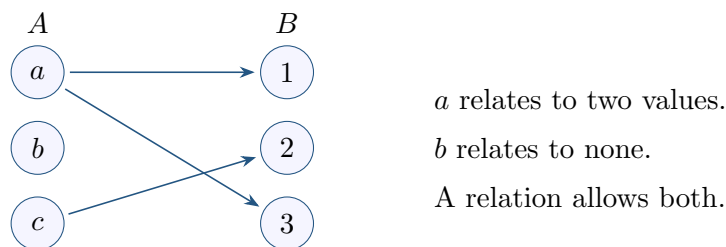
$$\begin{aligned} A \times B = \{ & (a, 1), (a, 2), (a, 3), \\ & (b, 1), (b, 2), (b, 3), \\ & (c, 1), (c, 2), (c, 3) \}. \end{aligned}$$

A **binary relation from A to B** is a subset

$$R \subseteq A \times B.$$

The notation xRy is shorthand for $(x, y) \in R$. We read it as “ x is related to y .” For example, take

$$R = \{(a, 1), (a, 3), (c, 2)\}.$$



The Cartesian product $A \times B$ contains all nine possible ordered pairs. A relation chooses any collection of those pairs. Thus every relation is a subset of $A \times B$, and conversely every subset of $A \times B$ defines a relation from A to B . Since a nine-element set has 2^9 subsets, there are $2^9 = 512$ relations from this A to this B .

1.1 Encoding a Relation by a Matrix

Once we fix an order for A and B , we can encode the relation in a **zero–one matrix**. Rows correspond to elements of A , columns correspond to elements of B , and

$$(M_R)_{xy} = \begin{cases} 1, & (x, y) \in R, \\ 0, & (x, y) \notin R. \end{cases}$$

For the relation above,

M_R	1	2	3
a	1	0	1
b	0	0	0
c	0	1	0

Each 1 records one pair in R . Reading the matrix backward recovers the relation, so the set of ordered pairs, the arrow picture, and the matrix all contain exactly the same information.

More generally, if $A = \{a_1, \dots, a_m\}$ and $B = \{b_1, \dots, b_n\}$, every relation $R \subseteq A \times B$ is encoded by an $m \times n$ matrix with

$$(M_R)_{ij} = 1 \iff (a_i, b_j) \in R.$$

1.2 Functions Are Special Relations

A function $f : A \rightarrow B$, written as a relation in set-builder form, is

$$f = \{(x, y) \in A \times B : y = f(x)\}.$$

It is a special relation with one added property: every element of A occurs as the first coordinate of exactly one ordered pair in the relation. In the matrix, this says that every row contains exactly one 1. The example relation above is not a function: row a has two 1s, while row b has none.

For a short example with different domain and codomain, let

$$X = \{u, v, w\}, \quad Y = \{1, 2, 3, 4\},$$

and define

$$f = \{(u, 2), (v, 4), (w, 2)\} \subseteq X \times Y.$$

Its matrix is

M_f	1	2	3	4
u	0	1	0	0
v	0	0	0	1
w	0	1	0	0

Every row has exactly one 1, so every input has exactly one output. A column may contain several 1s or none: both u and w correspond to 2, and no input corresponds to 1 or 3.

2 Relations on One Set and Graphs

A relation **on** V is a relation from V to itself:

$$R \subseteq V \times V.$$

For a first example, define a relation on $\mathbb{Z}$ by

$$xRy \iff x > y.$$

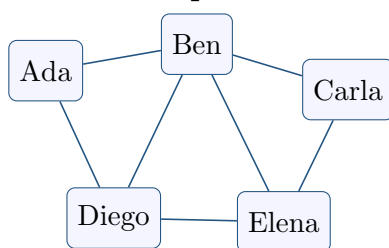
Then $5R2$ and $0R(-3)$, but $2R5$ does not hold. Other familiar relations on one set include equality, $\leq$, “has the same parity as,” and “is within distance one of.”

2.1 Graphs as Pictures of Relations

A **graph** consists of vertices and edges connecting them. The vertices are the objects under discussion; an edge records one instance of a relation.

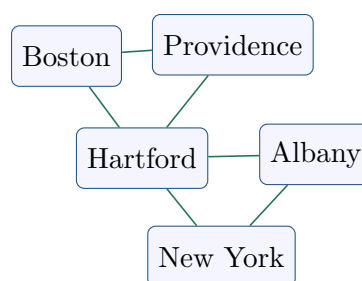
In an **undirected graph**, an edge has no direction. It is a natural picture for a symmetric connection: friendship, two cities joined by a road, or two computers connected on a network. An undirected edge between u and v corresponds to both ordered pairs (u, v) and (v, u) when we encode it as a relation.

Friendship network



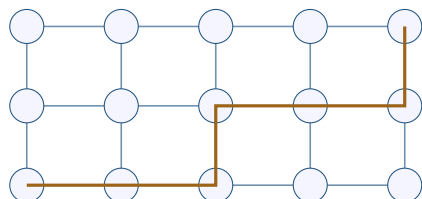
An edge means “are friends.”

Direct roads



An edge means “have a direct road.”

A grid as a graph. A street grid gives another natural example. Treat each intersection as a vertex and each street segment between neighboring intersections as an edge.



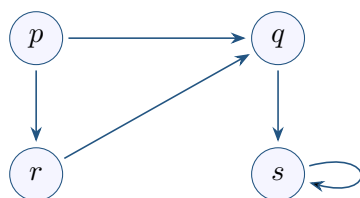
Vertices: intersections

Edges: street segments

Gold: one possible route

The drawing remembers which intersections are adjacent. The exact distances and angles are not part of the graph unless we add them as extra data.

In a **directed graph**, an edge is an arrow. This is useful when order matters: one person follows another, a flight goes from one airport to another, or one webpage links to another. A directed edge $u \rightarrow v$ represents the ordered pair (u, v) .



Each arrow $x \rightarrow y$ records (x, y) .

The loop records (s, s) .

Reversing an arrow changes the pair.

For the displayed relation on $V = \{p, q, r, s\}$,

$$R = \{(p, q), (p, r), (r, q), (q, s), (s, s)\}.$$

The loop at s records (s, s) . Its matrix, in the order p, q, r, s , is

$$M_R = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Thus a finite relation on one set has three equivalent representations:

$$\boxed{\text{ordered pairs}} \quad \longleftrightarrow \quad \boxed{\text{directed graph}} \quad \longleftrightarrow \quad \boxed{\text{zero-one matrix}}.$$

We will return to graphs later in the course and study paths, connectivity, trees, and other graph structures in more depth.

3 Example: Divisibility

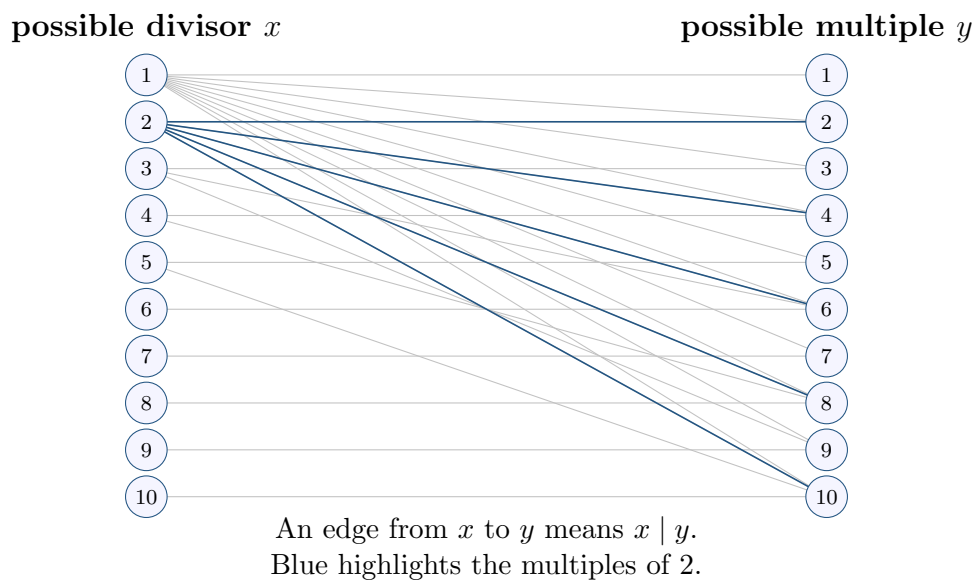
Let $D = \{1, 2, \dots, 10\}$. Define a relation on D by

$$xRy \iff x \mid y,$$

meaning that $y = xk$ for some integer k . For example, $2R8$ and $3R9$, while $4R6$ does not hold.

A map of the relation.

Although the same set D appears on both sides, we may draw two copies of it: possible divisors on the left and possible multiples on the right. Draw an edge from x to y exactly when $x \mid y$.



This picture emphasizes that divisibility is a subset of $D \times D$. The edge from the left-hand 2 to the right-hand 8 represents $(2, 8)$. Each number divides itself, so each left-hand x is joined to the right-hand copy of x .

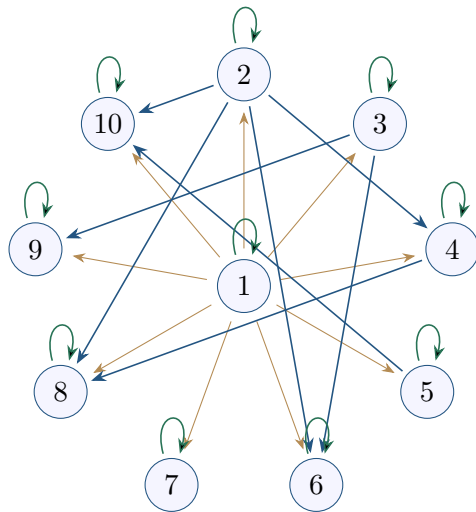
The matrix picture.

With rows and columns ordered $1, 2, \dots, 10$, the same relation is

M_R	1	2	3	4	5	6	7	8	9	10
1	1	1	1	1	1	1	1	1	1	1
2	0	1	0	1	0	1	0	1	0	1
3	0	0	1	0	0	1	0	0	1	0
4	0	0	0	1	0	0	0	1	0	0
5	0	0	0	0	1	0	0	0	0	1
6	0	0	0	0	0	1	0	0	0	0
7	0	0	0	0	0	0	1	0	0	0
8	0	0	0	0	0	0	0	1	0	0
9	0	0	0	0	0	0	0	0	1	0
10	0	0	0	0	0	0	0	0	0	1

The first row is all 1s because 1 divides every element of D . The diagonal is all 1s because every positive integer divides itself.

The directed graph picture. We can instead use one vertex for each element of D and draw the arrow $x \rightarrow y$ when $x \mid y$.



Gold: 1 divides every other number.

Blue: other proper divisibility arrows.

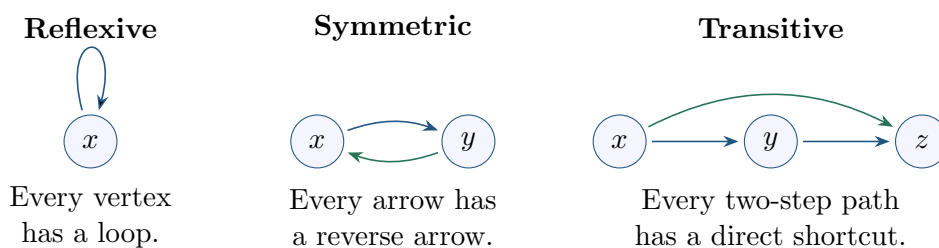
Green loops: $x \mid x$.

Green loops record $x \mid x$. The remaining arrows record proper divisibility. For example, there are arrows $2 \rightarrow 4$, $2 \rightarrow 6$, $2 \rightarrow 8$, and $2 \rightarrow 10$.

4 Three Properties of Relations

Let R be a relation on a set A .

- R is **reflexive** if $\forall x \in A, xRx$. Every vertex has a loop.
- R is **symmetric** if $\forall x, y \in A, xRy \implies yRx$. Every arrow has a reverse arrow.
- R is **transitive** if $\forall x, y, z \in A, (xRy \wedge yRz) \implies xRz$. Every two-step path has a direct shortcut.



The variables need not be distinct. To disprove one of these universal properties, it is enough to find a single counterexample.

Relation on $\mathbb{Z}$	Reflexive	Symmetric	Transitive
xRy iff $x = y$	Yes	Yes	Yes
xRy iff $x \leq y$	Yes	No	Yes
xRy iff $x < y$	No	No	Yes
xRy iff $ x - y \leq 1$	Yes	Yes	No
xRy iff $x - y$ is even	Yes	Yes	Yes

Examples of failure. The relation $\leq$ is not symmetric because $1 \leq 2$ but $2 \not\leq 1$. The relation $<$ is not reflexive because $0 < 0$ is false. The distance relation is not transitive: $0R1$ and $1R2$, but $0R2$ does not hold.

Reading the matrix. Reflexivity means every diagonal entry is 1. Symmetry means $M_R = M_R^T$. Transitivity means that whenever entries (i, j) and (j, k) are 1, entry (i, k) is also 1.

The divisibility relation on D is reflexive and transitive. It is not symmetric: $2 \mid 4$, but $4 \nmid 2$.

5 Equivalence Relations and Partitions

An **equivalence relation** is a relation that is reflexive, symmetric, and transitive. It captures the idea that two objects are the same according to a chosen feature.

When an equivalence relation E is written using the symbol $\sim$, the notation $x \sim y$ means $(x, y) \in E$. Thus $x \sim y$ has the same meaning as xRy : in each case, the ordered pair (x, y) belongs to the relation. Only the symbol used to name the relation has changed.

A **partition** of a set A is a collection of nonempty subsets, called **blocks**, such that every element of A lies in exactly one block. Equivalently, the blocks are pairwise disjoint and their union is A .

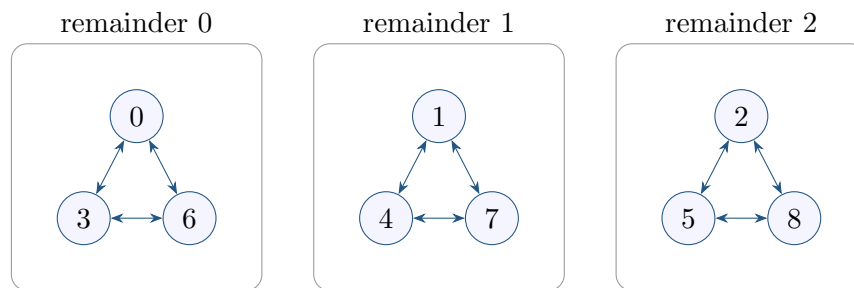
5.1 Congruence Modulo Three

On $D_9 = \{0, 1, \dots, 8\}$, define

$$x \sim y \iff 3 \mid (x - y).$$

The relation is reflexive, symmetric, and transitive, so it is an equivalence relation. It partitions D_9 into the three blocks

$$C_0 = \{0, 3, 6\}, \quad C_1 = \{1, 4, 7\}, \quad C_2 = \{2, 5, 8\}.$$



Every pair within a class is related. Self-loops are omitted here.

The equivalence class of x is

$$[x] = \{y \in D_9 : x \sim y\}.$$

Thus $[0] = C_0$, $[1] = [4] = C_1$, and $[2] = C_2$. Different representatives may name the same class.

In general, the equivalence classes of an equivalence relation form a partition. Reflexivity puts each x in $[x]$, so the classes cover the set. If $[x]$ and $[y]$ share an element z , then xRz and yRz . Symmetry gives zRy , and transitivity gives xRy . It follows that $[x] = [y]$. Thus two classes are either equal or disjoint.

For $i \geq 1$, let J_i denote the $i \times i$ matrix in which every entry is 1.

Lemma 1. *Let R be an equivalence relation on a finite set A , and let C be an equivalence class containing i elements. The submatrix of M_R whose rows and columns are indexed by the elements of C is J_i .*

Proof. Choose $a \in A$ such that $C = [a]$. If $x, y \in C$, then aRx and aRy . Symmetry gives xRa , and transitivity then gives xRy . Therefore every ordered pair in $C \times C$ belongs to R , so every entry in the submatrix is 1. The submatrix is J_i . $\square$

5.2 The Matrix before and after Reordering

First use the original order 0, 1, 2, 3, 4, 5, 6, 7, 8. Members of the three classes are interleaved, so the pattern of 1s is interleaved too:

$M_\sim$	0	1	2	3	4	5	6	7	8
0	1	0	0	1	0	0	1	0	0
1	0	1	0	0	1	0	0	1	0
2	0	0	1	0	0	1	0	0	1
3	1	0	0	1	0	0	1	0	0
4	0	1	0	0	1	0	0	1	0
5	0	0	1	0	0	1	0	0	1
6	1	0	0	1	0	0	1	0	0
7	0	1	0	0	1	0	0	1	0
8	0	0	1	0	0	1	0	0	1

Now reorder the elements by equivalence class:

$$0, 3, 6, \quad 1, 4, 7, \quad 2, 5, 8.$$

The same relation now has the matrix

	0	3	6	1	4	7	2	5	8
0	1	1	1	0	0	0	0	0	0
3	1	1	1	0	0	0	0	0	0
6	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	1	0	0	0
4	0	0	0	1	1	1	0	0	0
7	0	0	0	1	1	1	0	0	0
2	0	0	0	0	0	0	1	1	1
5	0	0	0	0	0	0	1	1	1
8	0	0	0	0	0	0	1	1	1

For example, the all-ones block contributed by any of the three equivalence classes is

$$J_3 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

Using this notation, the reordered matrix has the compressed form

$$M_\sim = \begin{pmatrix} J_3 & 0 & 0 \\ 0 & J_3 & 0 \\ 0 & 0 & J_3 \end{pmatrix} = \text{diag}(J_3, J_3, J_3).$$

Each J_3 records all ordered pairs inside one equivalence class. Each zero block says that elements from different classes are not related.

6 The Block-Matrix Characterization

The lemma and the previous example lead to a general theorem.

Claim 1. *Let R be a relation on a finite set A . Then R is an equivalence relation if and only if the elements of A can be reordered so that*

$$M_R = \text{diag}(J_{n_1}, J_{n_2}, \dots, J_{n_t})$$

for positive integers $n_1 + \dots + n_t = |A|$. All off-diagonal blocks are zero matrices.

Proof. Forward direction. Suppose R is an equivalence relation. Its equivalence classes form a partition

$$A = C_1 \sqcup C_2 \sqcup \dots \sqcup C_t.$$

List all elements of C_1 first, then all elements of C_2 , and so on. By Lemma 1, the matrix block for $C_i \times C_i$ is $J_{|C_i|}$.

If $x \in C_i$ and $y \in C_j$ with $i \neq j$, then xRy would imply $[x] = [y]$, contradicting that the classes are distinct. Hence every entry between different classes is 0. Therefore the reordered matrix has all-ones blocks on the diagonal and zero blocks elsewhere.

Converse direction. Suppose the elements can be ordered so that M_R has this block-diagonal form. Let C_i be the elements indexing the block J_{n_i} . Every element is related to itself because all diagonal entries are 1, so R is reflexive.

If xRy , then the entry for (x, y) is 1, so x and y lie in the same block. That block is all ones, so the entry for (y, x) is also 1. Thus R is symmetric.

Finally, suppose xRy and yRz . The first relation puts x and y in one block, and the second puts y and z in one block. Because the blocks are disjoint, all three elements lie in the same block. That block is all ones, so xRz . Thus R is transitive.

Therefore R is an equivalence relation. □

The theorem gives three equivalent ways to recognize the same finite structure:

